

Gauging Time Reversal Symmetry in Quantum Gravity: Arrow of Time from Spontaneous Symmetry Breaking

Deepak Vaid*

Department of Physics, National Institute of Technology Karnataka (NITK), India
and

Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India

* dvaid79@gmail.com

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Discuss
work by
Yoshida on
the stabil-
ity of SPT
states at
non-zero
tempera-
ture and
possible
relation to
emergence
of stable
geometry.
[1, 2]

Abstract

The question of the origin time's arrow is a major outstanding problem in physics. Here we present a possible mechanism for generation of a cosmological arrow via spontaneous symmetry breaking in a theory of quantum gravity. In [3], Chen and Vishwanath have shown that a local notion of time-reversal symmetry can be encoded into a gauge field living on a tensor network. Using the fact that tensor networks can be used to describe the state space of Loop Quantum Gravity [4, 5] we show that a spontaneous symmetry breaking mechanism operating on such tensor networks can lead to the generation of an arrow of time corresponding to the generation of non-zero order parameter for the Chen-Vishwanath gauge field.

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1 Introduction: Time's Arrow and Quantum Gravity

The question of “time” has perplexed humanity since the dawn of the civilization. Modern physics provides two conflicting roles for time. In quantum mechanics, time is simply an external parameter representing external clocks *w.r.t.* which the evolution of a quantum system takes places. In general relativity, on the other hand, time is a dimension on par with the three spatial dimensions and taken together, the four dimensions constitute a dynamical entity known as “spacetime” which serves as the arena on which all physics processes take place. However, unified though it may be in a certain geometrical sense with the spatial dimensions, time retains a character distinct from them. The Universe we live in appears to have a distinct *arrow of time*¹, a sense in which time *flows* from the “past” and towards the “future”. Spatial dimensions do not appear to display such behavior, with all directions being “equally likely” - in the sense that the matter distribution in the Universe does not appear to have a preferred spatial orientation on the largest observable scales².

The equations of motion of physical systems, in classical and quantum mechanics and in general relativity are manifestly time-symmetric [8, 9]. Yet, we know very well that macroscopic systems *appear* to be endowed with a preferred direction for time evolution. In classical physics this is commonly understood as a consequence of the thermodynamic second law for systems with large number of degrees of freedom. According to the second law, *isolated systems* always evolve from a state with low entropy to a state with high entropy. Thus the direction of increasing entropy can be identified with the direction of the flow of time. In quantum mechanics the situation was unclear until recently when advances in the field of many body localization and the eigenstate thermalization hypothesis [10, 11, 12, 13] have clarified that many body quantum systems coupled to an environment with a sufficiently large Hilbert space will evolve such that their final state is the equilibrium or highest entropy state, regardless of the initial starting state of the combined system. In other words even if the Universe started in a pure state, individual subsystems within the Universe would *almost* always evolve from states of low entropy to states of high entropy. There are also several proposals for how an arrow of time can spontaneously arise even in closed systems [14, 15, 16, 17]³. The above considerations, however, apply only to closed systems. The universe is not, to the best of our knowledge, a closed system.

Our universe appears to have a well-defined arrow of time on the largest cosmological scales. In particular, the initial state of the universe - at the so-called “Big-Bang” - appears to have been a low entropy state and over “time” the overall entropy of the Universe has increased and more and more complex structures have formed leading eventually to the present day epoch where systems have become complex enough to allow for the emergence of life⁴. One could argue that from the perspective of Popescu, Short and Winter [10] that the emergence of an arrow of time is inevitable in any subsystem of the Universe and that any such system coupled to a sufficiently large environment will always evolve towards a thermal state even if the combined system - *i.e.* the

¹see [6] for a comprehensive review of the arrow of time in classical, quantum and gravitational contexts.

²Whether or not the Universe is spatially isotropic on the largest scales is a matter of some debate [7].

³see, however, [18, 19, 20] for comments on and alternatives to the proposed resolution of the arrow of time given in [14]

⁴One should, in general, be careful not to confuse the two separate issue of “complexity” and “entropy”, though they clearly are intertwined in many cases. Entropy is commonly associated with “disorder” and complex systems are typically thought of as being “ordered”, low entropy configurations. However, as Jeremy England has shown [21], in open systems which are exposed to a constant energy flux, the formation of complex self-replicating structures which dissipate energy efficiently is inevitable. From this perspective complexity and entropy are mutually competing aspects of any system far from equilibrium.

Universe - starts in a pure state. The question is, to what extent does this apply to the Universe as a whole? Our Universe, on the largest observable scales, appears to exhibit a uniform sense of the flow of time. Individual isolated subsystems of the Universe might evolve towards a thermal state, but there is no reason, *a priori*, for why the sense of time-evolution in these different subsystems should be the same. This raises the question why does the observable Universe not consist of regions where time runs in opposite directions?

Ultimately, any question about the arrow of time must find their resolution in a theory of quantum gravity - the long sought after unification of quantum mechanics and general relativity which would permit us to understand the structure of spacetime at the Planck scale ($l_p \sim 10^{-34}m$) and in regions of high matter density and strong gravitational fields such as in and around black holes and near the Big Bang. One proposed route to quantum gravity is known as Loop Quantum Gravity ([22]) in which states of quantum geometry are modeled by graphs whose edges are labeled by representations of $SU(2)$. For obvious reasons, such graphs are also known as “spin networks”. In the framework of LQG several authors have discussed the role of time-reversal symmetry [23, 24] and how one could possibly detect an “anti-spacetime” - a region of spacetime where the local arrow of time points in the opposite direction to that of the overall global arrow in that spacetime.

Several authors [4, 5, 25] have shown that there exists a mapping between spin-networks and “tensor networks” which are a tool used to study the properties of quantum spacetime in approaches to quantum gravity inspired by the AdS/CFT correspondence. In [3] Chen and Vishwanath have shown how time-reversal symmetry can be made from a global to a local symmetry and described its action on tensor networks states.

In this work we build upon these developments to present a possible resolution of the question of the origin of a cosmological arrow of time by introducing a Z_2 gauge field representing a local time-reversal symmetry living on spin-networks. We argue that symmetry breaking of this gauge field will lead to the emergence of an arrow of time in semiclassical states of geometry and will also lead to the emergence of gapless excitations - Dirac fermions - which could be interpreted as matter degrees of freedom. This microscopic approach is complemented by prior work [23, 24] which permits a macroscopic, continuum description of the Z_2 gauge field as a symmetry which changes the sign of the determinant of the tetrad field.

The layout of this paper is as follows. In [section 2](#) we explain the crucial differences between Symmetry Protected Topological (SPT) order, Topological Order (TO) and Symmetry Enriched Topological (SET) order and outline why we expect our classical spacetime to arise as a SPT phase of some underlying fundamental theory. In [section 3](#) we explain the concept of matrix product states and tensor network states and show how *local* time reversal symmetry can be implemented in tensor network states (TNS). In [section 4](#) the correspondence between spin-network states of LQG and tensor networks is elaborated. In [section 5](#) we introduce the volume operator of LQG and explain how in the simplest case - when all spin network edges are labeled by $j = 1/2$ representation of $\mathfrak{su}(2)$ - the *local* action of time-reversal symmetry causes the volume of a given region of space to change sign. At the level of the classical Palatini action such an effect can be modeled by introducing a scalar field which couples non-trivially to the tetrad and possesses a local Z_2 symmetry. In [section 6](#) we discuss the CZX model which describes a SPT phase with gapless edge excitations in a many body system and show how spin-network can be mapped to the CZX model. This correspondence is then used to argue for the emergence of an arrow of time from spontaneous symmetry breaking in [section 7](#). Finally we end with a discussion of our results, address possible criticisms and suggest problems for future research in [section 9](#). For those not from the quantum gravity community necessary background on LQG and spin-networks states is provided in [Appendix B](#) and [Appendix D](#). Time reversal in the context of the connection formulation in classical GR is discussed in [Appendix C](#). For completeness, in [Appendix A](#) we give a brief outline of how time-reversal symmetry is implemented in quantum mechanical systems.

2 Topological Order and Spacetime Geometry

There exist many materials in nature which display transitions between phases which cannot be identified by measuring the change in the expectation value of some *local* order parameter field.

These systems fall outside the Landau-Ginzburg paradigm for classifying the phase space of many body systems. The order (or disorder) in these systems is characterized by the expectation values of *non-local* operators such as the Wilson loop operator:

$$\langle \mathcal{O}_\gamma \rangle = \langle \Psi | \exp \left(-i \oint_\gamma dx^\mu \mathbf{A}_\mu \right) | \Psi \rangle \quad (1)$$

where γ is a closed path. Such phases are characterized by expectation values of topological operators. By now, a bewildering variety of topological phases have been discovered in condensed matter systems and sometimes it gets difficult to keep track of the terminology associated with the different phases. Senthil [26] provides an accessible description of some of the common types of topological phases as shown below. This listing is not exhaustive but is indicative of the diversity of different types of topological orders.

Species	Description	Examples
SPTs - Symmetry Protected Topological states	Short-range entangled states, which can be deformed to a trivial product state using local unitary transformations, at the cost of breaking a protected symmetry. In other words the non-trivial topological order of the state is <i>protected</i> by the existence of an unbroken symmetry.	Topological Insulators, AKLT State, Haldane phase
TOs - Topological Orders	Long-range entangled states which cannot be transformed into a trivial product state by local unitary transformations even by breaking all global symmetries	FQHE, Spin Liquids
SETs - Symmetry Enriched Topological States	These are TOs, which in addition to long-range entanglement, also have some global symmetry.	FQHE, Spin Liquids

Figure 1: Topological orders: terminology, characteristics and examples

Significant conceptual and technical breakthroughs [27, 28] in the last decade lead us to believe that the geometry of spacetime arises from the entanglement between the various subsystems of some underlying pre-geometric strata. As first argued ⁵ in a simple and elegant fashion by Van Raamsdonk in 2010 [27], the amount of entanglement between two different regions of spacetime determines the geometric connectivity of the points within those regions with each other. This argument builds upon the earlier groundbreaking work of Ryu and Takayanagi [30, 31] who showed that the entropy of entanglement between two different regions of spacetime - in the restricted setting of AdS spacetime - is proportional to the area of the *minimal* surface which separates the two regions:

$$S_{EE} \propto A_{min}. \quad (2)$$

Now, imagine disconnecting the two bulk regions from each other, by “pinching” off the boundary surface joining the two region. As the area of the surface shrinks to zero, points in one region become increasingly disconnected from points in the other, until when the area reaches zero, the two regions become completely geometrically separate. At this stage, as per the RT formula (2) the entanglement entropy between the two regions also goes to zero. This simple argument - anchored in the well understood physics of the AdS/CFT correspondence - is sufficient to suggest that an apparently smooth, extended macroscopic geometry arises when smaller regions of quantum spacetime are “stitched” together by entanglement.

There is a flip side to this argument. If the metric distance between two neighboring regions of space is inversely proportional to the entanglement entropy between the two regions, then as the

⁵In earlier work [29] Swingle had suggested that the emergence of spacetime could be seen as a consequence of applying the renormalization group to the tensor network representation of the quantum state of a many body system. While this work was deeply significant in and of itself, it was van Raamsdonk who first explained in a clear manner how the precise connection between geometry and entanglement should arise.

entanglement entropy between any two regions increases the metric distance between the two should decrease. Now, the spacetime we observe around us exhibits the property of *locality*. The metric geometry changes smoothly as we increase the scales and distances between different systems. For instance, we “know” that Saturn is closer to Earth than is Alpha Centauri. This means that the geometric degrees of freedom around our planet have a greater amount of entanglement with the degrees of freedom around Saturn than they do with the degrees of freedom around Alpha Centauri. If this were not the case, then it would not be possible to have a *sense* of the separation between the Earth and Alpha Centauri.

The same structure is observed on the largest scales in the universe. This suggests that the underlying many-body system from which our macroscopic geometry emerges exhibits short range entanglement (SRE) rather than long range entanglement (LRE). In a state with LRE the macroscopic geometry would be ill-defined as points arbitrarily far apart from each other with respect to the microscopic geometry could be arbitrarily close to each other with respect to the emergent macroscopic geometry. This leads us to the following ansatz:

Emergent macroscopic geometry exhibits (approximate) smoothness and locality when the underlying quantum many body system is in a SRE phase.

However, only SRE is not sufficient to ensure the emergence of a connected, bulk geometry. This is because states with SRE can, in general, be deformed to trivial product states by local unitary operations. In order to have a state with *non-trivial* topological order which cannot be deformed into trivial product states by local unitaries the existence of an unbroken symmetry is required. As long as the symmetry is preserved the SRE state will have non-trivial topological order and cannot be deformed into a trivial state. Such a state is therefore said to possess *symmetry protected topological* (SPT) order. In light of this fact the statement of the previous ansatz is modified as follows:

Emergent macroscopic geometry exhibits (approximate) smoothness and locality when the underlying quantum many body system is in a SPT phase.

3 Matrix Product States and All That

Consider a many body system consisting of N quantum degrees of freedom. The most general many body state of such a system can be written as:

$$|\psi\rangle = \sum_{i_1, i_2, \dots, i_N} C_{i_1, i_2, \dots, i_N} |i_1, i_2, \dots, i_N\rangle \quad (3)$$

where $|i_k\rangle$ is the state of the k^{th} degree of freedom. For instance, in a spin-chain, the single body Hilbert space would be $\mathcal{H}^2 = \{|\uparrow\rangle, |\downarrow\rangle\}$ and the sum in the above equation would be over all the possible spin configurations of the sites in the chain. It is easy to see that the total Hilbert space of the spin-chain is:

$$\mathcal{H}^{2^N} = \bigotimes_{i=1 \dots N} \mathcal{H}_i^2 \quad (4)$$

and the dimensionality of this Hilbert space is 2^N . As N increases, the amount of information required to specify a given many-body state increases exponentially. Moreover in the representation (3), there is no way to distinguish short-range correlations from long-range ones. The co-efficients $C_{i_1, i_2, \dots, i_N}$ are defined over the entire Hilbert space and cannot be decomposed into local contributions in any simple manner.

Most importantly for our purposes, it is not clear what the local action of the time-reversal symmetry would be on a state of the form (3). One can define the global action:

$$\mathcal{T}|\psi\rangle = \sum_{i_1, i_2, \dots, i_N} C_{i_1, i_2, \dots, i_N}^* U_1 \otimes \dots \otimes U_N |i_1, i_2, \dots, i_N\rangle \quad (5)$$

Here the action of $\mathcal{T}$ on a state is expressed as a product of a unitary transformation U followed by complex conjugation K : $\mathcal{T} = KU$. Each factor of U_i acts on the i^{th} site, while complex conjugation

acts on the coefficients $C_{i_1, i_2, \dots, i_N}^*$. It is precisely this complex conjugation which makes it difficult to understand how to apply time reversal *locally*, *i.e.* to a subset of the system $\{i_m, \dots, i_{m+n}\}$, rather than to the whole system. What is needed is a way to express (5) in a way that allows us to act with the complex conjugation *locally*, and only on select parts of the system, rather than on the global state. It turns out it is possible to express (5) in such a way using the technique of matrix product states (MPS) or their generalization to dimensions greater than one, tensor network states (TNS).

3.1 Matrix Product States

A typical MPS is written as [32, Sec 3]:

$$|\psi\rangle = \sum_{i_1, i_2, \dots, i_N} \text{Tr} \left[A_{i_1}^{(1)} A_{i_2}^{(2)} \dots A_{i_N}^{(N)} \right] |i_1, i_2, \dots, i_N\rangle \quad (6)$$

where $A_{i_k; \alpha\beta}^{(k)}$ is a set of matrices associated with the k^{th} lattice site; with $i_k \in \{1, 2, \dots, D\}$, where D is the dimension of the single site Hilbert space and $\alpha, \beta \in \{1, 2, \dots, d\}$, where d is known as the “bond dimension”. As an example, for a chain of spin 1/2 particles, each site can be one of two states $\{|\uparrow\rangle, |\downarrow\rangle\}$ and thus with each site we associate two matrices: $A_{\uparrow; \alpha\beta}^{(k)}$ and $A_{\downarrow; \alpha\beta}^{(k)}$. From now on for ease of notation, instead of using the up/down notation $\{|\uparrow\rangle, |\downarrow\rangle\}$ we will use $\{|1\rangle, |0\rangle\}$ to represent the two states of a spin 1/2 system. Typically, we will consider systems invariant under translations, which allows us to use a single set of matrices $A_{i_k; \alpha\beta}$ for each site. We will do so in the following examples.

Working with the spin half chain, consider, for instance the matrices:

$$A_0 = (1); \quad A_1 = (0) \quad (7)$$

Inserting these into the MPS expression (6) - and keeping in mind that the subscripts $\{0, 1\}$ on the matrices correspond to the allowed states $\{|1\rangle, |0\rangle\}$ at each site, rather than to the sites themselves - we can see that whenever the state at a given site is $|1\rangle$, the coefficient of that term will vanish. Thus we are left with the result that (7) represents the simple state: $|00\dots 0\rangle$. A less trivial example is that of the GHZ (Greenberger-Horne-Zeilinger) [33] state:

$$|\psi_{GHZ}\rangle = |00\dots 0\rangle + |11\dots 1\rangle \quad (8)$$

for which the associated MPS matrices are:

$$A_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}; \quad A_1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad (9)$$

One can explicitly evaluate the expression (6) with the choice of above matrices for, say, a two-site system to find that it does indeed correspond to (8). It is important at this point to mention that it has been proven in [34] that *any* many-body state can be written in the form (6) as long as the bond dimension d is large enough. Therefore the MPS ansatz can be used to study local application of time-reversal, among other, symmetries to an arbitrary many-body system.

There is an illuminating diagrammatic notation [32] for MPS. The key component of this notation is the vertex diagram shown in Figure 2. The node labelled i represents the physical degree of freedom corresponding to the state $|i\rangle$ in the single site Hilbert space. The site index itself is not shown.

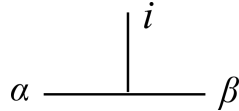


Figure 2: Single vertex of a matrix product state. Internal indices are denoted by α, β . Physical indices by $i, j, \dots$

Figure 3 shows how two connect two such vertices. In terms of matrices, such a connection corresponds to multiplying two site matrices: $A_{i_1; \alpha\gamma}^1 A_{i_2; \gamma\beta}^2$ (where the sum over the repeated index

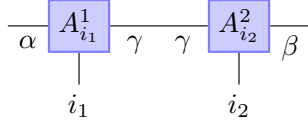


Figure 3: A connection between two adjacent vertices corresponds to multiplying the associated matrices.

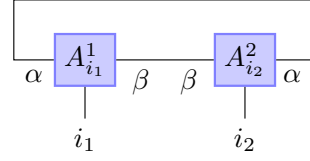


Figure 4: To take the trace we complete the loop by connecting the first and last vertices

γ is implied). In order to obtain the numerical value of the wavefunction coefficient in (6), the trace has to be taken after connecting all the vertices together. This operation is illustrated in Figure 4, by the line connecting $A_{i_1}^1$ and $A_{i_2}^2$ from the outside.

An illustration of use of this notation to represent a MPS is shown in Figure 5 for a system with five spins.

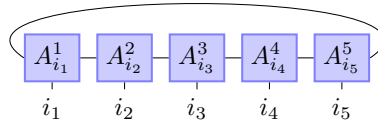


Figure 5: Diagrammatic representation of a matrix product state (MPS). The matrix A_i lives at the i^{th} lattice site. Indices $i_1, \dots, i_5$ represent the *physical* degrees of freedom at each site. The trace operation is represented by the line connecting by the first and last sites (figure courtesy of [32])

3.2 Tensor Network States

One can easily extend this notation to arbitrary lattices in arbitrary number of dimensions. When dealing with graphs with degree greater than two, the term “tensor network states” or “tensor product states” is used instead of “matrix product states”, because each matrix now has more than two internal indices. An example is shown in Figure 6 which shows the diagrammatic representation for the state for a system defined on a two-dimensional square lattice, where the physical degrees of freedom live on the edges. This state is defined by two sets of tensors: a four index object $A_{\alpha\beta\gamma\eta}$ for the vertices and a two index object $g_{i_e;\alpha\beta}^e$ for the edges. Since the physical degrees of freedom only live on the edges, only the edge tensor has the corresponding index i_e , where e represents the given edge and $i_e \in \{1, \dots, D\}$, with D , as before, being the dimension of the Hilbert space of the physical degree of freedom.

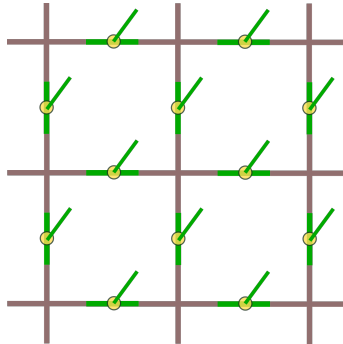


Figure 6: Tensor network state (TNS) representation of the toric code wavefunction. Each edge (green) has a physical index i and two internal indices α, β . The corresponding tensor is $g_{i_e;\alpha\beta}^e$ where e labels the edge and i_e labels the state of the edge. The vertices have only internal indices and the corresponding tensor is written as $A_{\alpha\beta\gamma\delta}^v$ where v labels the given vertex.

Number of degrees of freedom	N	
Site Hilbert space	$\mathcal{H}$	
Number of site degrees of freedom	$d = \dim \mathcal{H}$	
Bond dimension	$D \in \mathbb{N}$	
Vertex Tensors	$A_{i_v; \alpha_1 \dots \alpha_{v_k}}^v$	v : vertex i_v : vertex state/physical index $\alpha_1 \dots \alpha_{v_k} \in \{1, \dots, D\}$: internal (“bond”) indices
Edge Tensors	$g_{i_e; \alpha \beta}^e$	e : edge label i_e : edge state/physical index $\alpha, \beta \in \{1, \dots, D\}$: bonding indices

Table 1: Data required to specify the state of a tensor network

In general given an arbitrary graph Γ , the full wavefunction for such a graph is given by:

$$|\Psi\rangle = \sum_{\{i_1, \dots, i_{n_v}\}, \{j_1, \dots, j_{n_e}\}} \text{Tr} \left[\prod_{v \in \Gamma} A_{i_v}^v \prod_{e \in \Gamma} g_{j_e}^e \right] |i_1, \dots, i_{n_v}; j_1, \dots, j_{n_e}\rangle \quad (10)$$

Here n_v, n_e are the total number of vertices and edges in the graph respectively; i_k, j_k are the physical degrees of freedom for vertices and edges respectively. The products are over all vertices and edges of the graph and the sum is over all possible configurations of the physical degrees of freedom on the graph. We have suppressed the internal indices on the tensors for simplicity. In the next section we will see that (10) is the precise form of the expression for the wavefunction of a spin-network state in LQG with the only difference being that the physical (or “bulk”) degrees of freedom are located at the vertices instead of on the edges as in the case of the toric code.

3.3 Dangling Indices: Bulk vs Boundary Degrees of Freedom

We will also need to consider states with *dangling indices*, *i.e.* “internal” or bonding degrees of freedom which are not traced over. As we will see in the next section, spin-network states have two different types of such free indices. One kind comes from the intertwiner degrees of freedom which encode the bulk geometry of the spacetime and the second correspond to the boundary states of quantum geometry of a bulk region. Consider the MPS for a two-site system as shown in Figure 3. Such as state, before would be written as:

$$|\Psi_{\alpha\beta}\rangle = \sum_{\{i_1, i_2\}} A_{i_1; \alpha\gamma}^1 A_{i_2; \gamma\beta}^2 |i_1, i_2\rangle \quad (11)$$

Now the state is a two-index object. Likewise if the graph has n dangling edges, then the resulting state will also have n indices of size equal to the bond dimension. Consider a graph Γ , which has a subset of n vertices which are connected to only a single edge, each. These vertices and their accompanying edges $\{v_1, \dots, v_n; e_1, \dots, e_n\}$ form a subset $\partial\Gamma \subset \Gamma$. Let $\bar{\Gamma} = \Gamma/\partial\Gamma$, be the “bulk” of the graph which does not have any dangling edges. Resulting state is n index object.

$$|\Psi_{\alpha_1, \dots, \alpha_n}\rangle = |\Psi_{\text{bulk}}\rangle |\alpha_1, \dots, \alpha_n\rangle \quad (12)$$

where:

$$|\Psi_{\text{bulk}}\rangle = \sum_{\{i_v, j_e\} \in \bar{\Gamma}} \left[\prod_{v \in \bar{\Gamma}} A_{i_v}^v \prod_{e \in \bar{\Gamma}} g_{j_e}^e \right] |\{i_v\}, \{j_e\}\rangle \quad (13)$$

3.4 Global Gauge Symmetry of MPS

The central thrust of this work is the idea that gauge symmetry breaking in spin-network states can lead to the emergence of an arrow of time. For this to work, we have to understand how gauge

symmetry can act on MPS - and thereby on spin-network states. Now any MPS is given by data as shown in Table 1. Any such state has a *global* gauge symmetry under which all the matrices are transformed by a constant matrix, as in:

$$A_{i_k}^{(k)} \rightarrow M A_{i_k}^{(k)} M^{-1} \quad (14)$$

where M is a constant matrix. Then it is easy to see that the resulting state will be the same as before, because all factors of M, M^{-1} will cancel when taking the trace:

$$|\psi\rangle \rightarrow \sum_{i_1, i_2, \dots, i_N} \text{Tr} \left[M A_{i_1}^{(1)} M^{-1} M A_{i_2}^{(2)} M^{-1} \dots M A_{i_N}^{(N)} M^{-1} \right] |i_1, i_2, \dots, i_N\rangle = |\psi\rangle \quad (15)$$

Diagrammatically the effect of a such a gauge symmetry can be illustrated as follows:

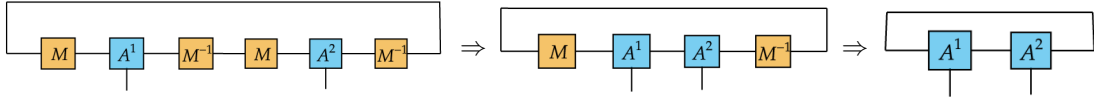


Figure 7: Action of a global gauge transformation on a matrix product state. After tracing out all the degrees of the freedom, the resulting state is unchanged.

3.5 Local Gauge Action

For our purposes what is needed is to be able to understand how to convert the global symmetry into a local one. We know how to do this for symmetries under groups such as $U(1)$ or $SU(2)$. For a system of spin, for instance, endowing the system with a global $SU(2)$ symmetry implies that the system should remain unchanged under a simultaneous rotations of *all* the spins under the same $SU(2)$ rotation. In order to make this into a local symmetry, we simply restrict the rotations to individual spins such that instead of a single element of $SU(2)$ there now exists a gauge *field* $g(\vec{x})$ (where $\vec{x}$ is the co-ordinate of individual sites either in a discrete or continuous system) which changes from point to point in the system. If we want to know how the state of a given spin changes we apply the value of $g(\vec{x})$ at the location of *that* spin only.

For time-reversal the situation becomes more subtle. This is because time-reversal is not a unitary operation, but an anti-unitary one, *i.e.* in addition to transformation of the degrees of freedom by some group element we must also take the complex conjugate of the state. For global time-reversal of a many body state of the form (3) this can be accomplished as follows:

$$\begin{aligned} \mathcal{T}|\psi\rangle &= \mathcal{T} \sum_{i_1, i_2, \dots, i_N} C_{i_1, i_2, \dots, i_N} |i_1, i_2, \dots, i_N\rangle \\ &= \sum_{i_1, i_2, \dots, i_N} C_{i_1, i_2, \dots, i_N}^* U_1 \otimes \dots \otimes U_N |i_1, i_2, \dots, i_N\rangle \end{aligned} \quad (16)$$

where the complex conjugation acts on the coefficient $C_{i_1, i_2, \dots, i_N}$ of the many-body state. It is clear that this is a global effect. It is not clear how to make complex conjugation into a *local* operation, at least as long as the state is written in the form (3). This is where the utility of the MPS or TNS representation becomes manifest. The action of (global) time-reversal on the MPS state (6) is given by:

$$\mathcal{T}|\psi\rangle = \sum_{i_1, i_2, \dots, i_N} \text{Tr} \left[A_{i_1}^{(1)} A_{i_2}^{(2)} \dots A_{i_N}^{(N)} \right]^* U_1 \otimes \dots \otimes U_N |i_1, i_2, \dots, i_N\rangle \quad (17)$$

From this expression one can see the effect of (global) time-reversal is equivalent to replacing the matrices in the above representation:

$$\tilde{A}_{i_k}^{(k)} = U^T_{i_k} A_{i_k}^{(k)*} \quad (18)$$

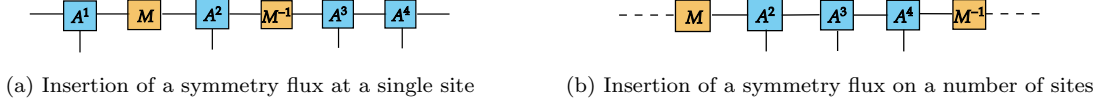


Figure 8: Insertion of local gauge symmetry fluxes in a MPS

Recall, that in the MPS representation (6) the expression $A_{i_k}^{(k)}$ is actually short-form for the full matrix: $A_{i_k; \alpha\beta}^{(k)}$. Here α, β are “bond” indices and connect matrices of neighboring sites with each other, k is the site index and $i_k \in \{1, 2, \dots, D\}$ indexes the basis states of the (physical) site Hilbert space. For simplicity ignoring all other sites we focus only on a single site in the MPS. The action of the unitary U on a single site k can then be expressed as⁶:

$$A_{i_k}^{(k)} |i_k\rangle \rightarrow \left(A_{i_k}^{(k)}\right)^* U_{i_k}^{j_k} |j_k\rangle = (U^T)_{j_k}^{i_k} \left(A_{i_k}^{(k)}\right)^* |j_k\rangle = \tilde{A}_{j_k}^{(k)} |j_k\rangle$$

with $\tilde{A}_{j_k}^{(k)}$ defined as in (18).

Now, for a time-reversal symmetric and short-range entangled (SRE) state the relation (18) can equivalently be written as [3, 35]:

$$\tilde{A}_{\alpha\beta}^{(k)} = M_{\alpha\gamma} A_{\gamma\eta}^{(k)} M_{\eta\beta}^{-1}, \quad (19)$$

where now the matrix M acts on the bond indices of A , rather than on the physical state index as in (18). This is illustrated below in Figure 8a. This can be used to extend the application of TRS to a group of neighboring sites rather than to only a single site. In this case each member of the neighborhood will undergo a transformation given by (19). The factors of M and M^{-1} which are located in the interior of the region will cancel each other out leaving only a factor of M and M^{-1} on either side of the boundary of the region as shown in Figure 8b.

We can extend this reasoning to two and three dimensional systems. In this case the site tensors will have three or more indices and a matrix insertion will be required on every leg. When applied to a subregion of a given lattice, the result will be that matrices on the interior will multiply to give the identity and we will be left with the non-trivial unitaries only on the boundary of the region as shown in Figure 9a.

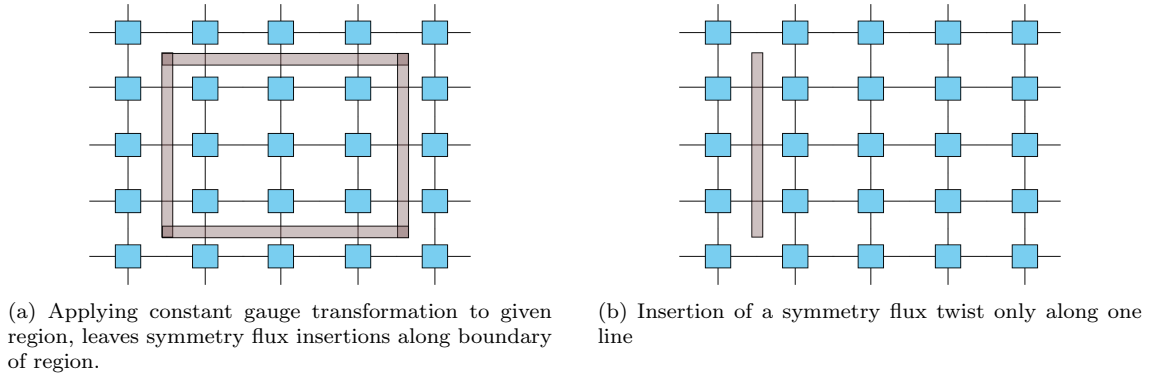


Figure 9: Local symmetry flux insertions in a 2D MPS

4 Spin Networks

Spin networks are very simple to describe physically. Consider an arbitrary graph Γ , as shown in Figure 10a, whose edges are labelled by representations $j_e \in \mathbb{Z}/2$ of $SU(2)$. This is very similar to

⁶The placement of the indices, lower and upper, on the unitary matrix U_i^j is used purely for notational elegance and has no physical significance here.

the tensor networks defined in the previous section, with the only difference being that now the edge Hilbert spaces can all be of different dimensionality $D_e = \dim[\mathcal{H}_e] = j_e(j_e + 1)$ depending on the representation j_e living on the respective edges. The tensor associated with each edge will then be a matrix $g_{i_e; \alpha\beta}^e$ of rank 2 and dimension D_e , with $i_e \in \{|-j_e\rangle, |-j_e + 1\rangle, \dots, |j_e - 1\rangle, |j_e\rangle\}$ being the state living on the edge⁷ taking values in $\mathcal{H}_e$.

To complete the definition of a spin-network we need to assign tensors $T_{j_v}^v$ to the vertices also. Now, as opposed to the edge tensors, the rank of the vertex tensors will vary, being given by the degree k_v of each vertex⁸. Given a vertex v , let $\{e_m^v; m \in \{1, \dots, k_v\}\}$ be the set of edges which meet at v . Then the tensor $T_{j_v}^v$ will be an element of:

$$\mathcal{H}_v = \left[\bigotimes_{m=1..k_v} \mathcal{H}_{e_m^v} \right] \quad (20)$$

which is the tensor product space of the edge Hilbert spaces $\mathcal{H}_{e_m^v}$. Later will we have to restrict further to the subspace $\mathcal{H}_v^{Inv}$ invariant under the action of $SU(2)$ gauge transformations.



(a) A graph with edges labelled by representations of $SU(2)$. Each j_k is an $SU(2)$ spin taking values in the set $\{1/2, 1, 3/2, 2, \dots\}$

(b) A single vertex connected to four edges, with each edge labeled by a representation j_i of $SU(2)$. The vertex is represented schematically by a tetrahedron. The spin on each edge can then be thought of as representing the normal to the correspond face of the tetrahedron.

Figure 10: States of quantum geometry represented by spin-networks

As a simple illustration of this consider the case of single vertex with four edges [Figure 10b](#). For simplicity, we set $j = 1/2$ for all edges. Now the state space $\mathcal{H}^2$ of a single edge is spanned by the states $\{|\uparrow\rangle, |\downarrow\rangle\}$, or equivalently $\{|1\rangle, |0\rangle\}$. The state space of all edges taken together is then: $\mathcal{H}^v = \{\mathcal{H}^2\}^{\otimes 4}$ and is spanned by states of the form $|m_1, m_2, m_3, m_4\rangle$ where the coefficients m_k take values in $\{0, 1\}$. An arbitrary state of the edge system can then be written as:

$$|\Psi_v\rangle = \sum_{\{m_k\}} C_{m_1, m_2, m_3, m_4} |m_1, m_2, m_3, m_4\rangle \quad (21)$$

where the sum is over all allowed spin states of the edges: $m_i \in \{-j_i, -j_i + 1, \dots, j_i - 1, j_i\}$ and C_{m_1, m_2, m_3, m_4} is the weight for the corresponding state vector. Not all states of the form (21) are physically allowed. We want quantum states which corresponds to classical states of three geometry. Physical states in the classical theory are required to satisfy the constraints discussed in the previous section. There are three constraints: the Hamiltonian constraint $\mathcal{H}$, the diffeomorphism constraint $\mathcal{C}^a$ and the Gauss constraint $\mathcal{G}^i$. We are interested in the integral form of the Gauss constraint (D.7b). When translated into the language of operators acting on spin-networks the *quantum* version of the Gauss constraint can be written as:

$$\sum_i J_i = 0 \quad (22)$$

where the Gaussian surface now encloses a single vertex of our spin-network and the quantum version of the Gauss constraint says that the sum of the angular momenta carried by the edge

⁷Recall that given a particle of spin j , the Hilbert space is spanned by a set of vectors $\{|j; m\rangle\}$, where $m \in \{-j, -j + 1, \dots, j - 1, j\}$. Here m is referred to as the “magnetic” quantum number. In the main text we have used just $|m\rangle$ for these states, rather than the full $|j; m\rangle$, for notational simplicity.

⁸In graph theoretical language, the “degree” a vertex is the number of edges which meet at that vertex.

spins should be zero. Physical states should satisfy this constraint. In other words:

$$(J_1 + J_2 + J_3 + J_4)|\Psi_v\rangle_{phys} = 0 \quad (23)$$

This relationship has a simple geometric interpretation. It is the statement that the faces of the tetrahedra in Figure 10b should close. Of course, (22) applies to all vertices, not just the 4-valent ones. Given a node with n edges attached to it, the resulting state is a quantum version of a classical polyhedron with n faces. The set of all face normals $\vec{N}_i$ must satisfy $\sum_i \vec{N}_i = 0$ if the polyhedron is to close. The angular momentum vector is the quantum analog of this normal vector and hence the classical closure condition:

$$\sum_i \vec{N}_i = 0 \quad (24)$$

implies the quantum closure condition (22).

Now as shown in Appendix D the wavefunction for a spin-network defined on an arbitrary graph can be written in the form:

$$|\Psi\rangle = \sum_{\{i_v\}, \{j_e\}} \text{Tr} \left[\prod_{v \in \Gamma} I_{i_v}^v \prod_{e \in \Gamma} D_{j_e}^e \right] |i_1, \dots, i_{n_v}; j_1, \dots, j_{n_e}\rangle \quad (25)$$

with the various terms in this equation being defined in Table 2.

Vertex Tensors	$I_{i_v; \alpha_1 \dots \alpha_{v_k}}^v$	v : vertex i_v : vertex state/physical index $\alpha_i \in \{1, \dots, d_i\}$; $d_i = 2j_i + 1$: contracts with the state on the i^{th} edge joined to the vertex.
Edge Tensors	$D_{j_e; \alpha\beta}^e$	e : edge j_e : $SU(2)$ label of edge $\alpha, \beta \in \{1, \dots, (2j_e + 1)\}$: bonding indices

Table 2: Data required for specifying the state of a spin network

We can see from the form of (25) and Table 2 that the expression for the state vector living on a spin network is identical to that of a tensor network state given in (10).

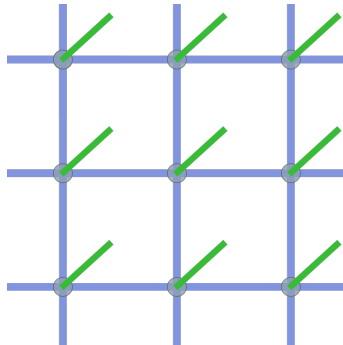


Figure 11: A spin-network as a tensor network whose “physical” degrees of freedom correspond to the allowed intertwiner states of four-valent vertices.

It is worth noting that the condition (23) is *exactly* the same as the “branching rules” for so-called string-net states as first described by Levin and Wen in [36].

4.1 Microscopic Structure of Spacetime

Spin-networks have a beautiful interpretation in terms of geometric observables. As such spin-networks are abstract graphs which do not require any background manifold. The whole idea is that the notion of a smooth manifold with some geometry *emerges* from considering superpositions or ensembles of large numbers of such graphs. The first step towards achieving this goal is to relate the physical variables living on the graph with geometric quantities. Since spin-networks arise as the quantization of Einstein's equations, which are themselves built out of purely geometric quantities such as the metric and Christoffel connection, it is natural that they would encode spacetime geometry. The manner in which this happens is as follows. Consider a single edge of a graph carrying a spin j . Whenever this edge intersects a surface, as shown in Figure 12a, it endows the surface with a quantum of area at the location of the puncture. The area arises as the eigenvalue of the Casimir operator $\hat{\mathbf{j}}^2$ acting on the state of the spin-network *at that edge*. These eigenvalues thus take the form:

$$A = 8\pi\gamma l_p^2 \sqrt{j(j+1)} \quad (26)$$

where l_p is the Planck length, γ is known as the Barbero-Immirzi parameter and j is the spin on that edge. From this equation it is clear that the area operator is gapped, *i.e.* it has smallest eigenvalue which is greater than zero. This occurs for the smallest value of spin $j = 1/2$,

$$A_{min} = 4\sqrt{3}\gamma l_p^2 \quad (27)$$

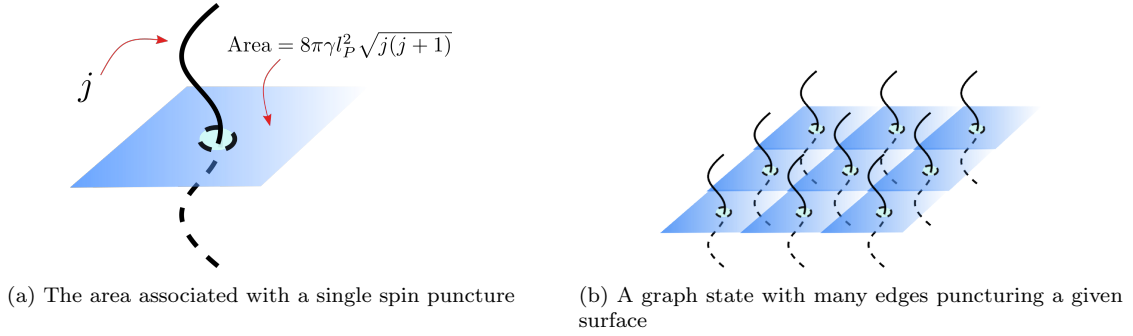


Figure 12: How macroscopic geometry emerges from quantum geometry

5 Volume Operator and Z_2 Symmetry

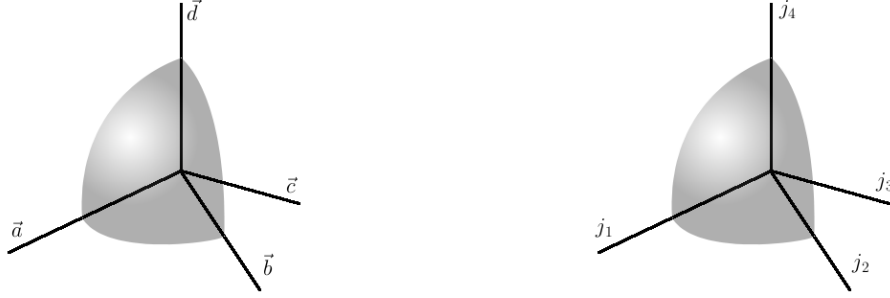
Having established the physical correspondence between spin-networks states and gauged tensor networks, we now return to the physics of the problem under consideration. We wish to argue that the *local* action of time-reversal symmetry on a spin network leads to a change in the sign of the eigenvalues of the LQG volume operator acting in that region. This effect can be modeled at the level of the Palatini action by introducing a Z_2 field ϕ which couples non-trivially to the tetrad.

There are two commonly used versions of the volume operator in LQG. They are named after the respective authors as the Rovelli-Smolín (RS) [37] or the Ashtekar-Lewandowski (AL) [38, 39] versions. These have the form:

$$\begin{aligned} \hat{V}_S^{RS} |\Psi_\Gamma\rangle &= \gamma^{3/2} l_p^3 \sum_{v \in S \cap \Gamma} \sum_{i,j,k} \left| \frac{i C_{reg}}{8} \epsilon^{ijk} \hat{J}_i \hat{J}_j \hat{J}_k \right|^{1/2} |\Psi_\Gamma\rangle \\ \hat{V}_S^{AL} |\Psi_\Gamma\rangle &= \gamma^{3/2} l_p^3 \sum_{v \in S \cap \Gamma} \left| \frac{i C_{reg}}{8} \epsilon_v(i,j,k) \epsilon^{ijk} \hat{J}_i \hat{J}_j \hat{J}_k \right|^{1/2} |\Psi_\Gamma\rangle \end{aligned} \quad (28)$$

where γ is the so-called Barbero-Immirzi parameter whose precise value or even physical relevance is a matter of some debate. Its actual value will not affect our arguments so for convenience we will set $\gamma = 1$; l_p is the Planck length; Γ is the spin-network; S is the subset of the graph whose

volume we wish to determine; $v \in S$ is the set of all vertices contained within S ; $\{\hat{J}_i\}$ are the angular momentum operators living on each edge attached to v .



(a) The volume associated with the solid angle spanned by three classical vectors is given by $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$

(b) The volume associated with the solid angle spanned by three quantum vectors is given by $\hat{J}_1 \cdot (\hat{J}_2 \times \hat{J}_3)$

Figure 13: Mapping classical volume to a quantum expression involves replacing classical vectors with angular momentum operators. Figure credit [22].

These two forms are identical except for the factor of $\epsilon_v(i, j, k)$ which occurs in the AL version. The $\epsilon^{ijk} \hat{J}_i \hat{J}_j \hat{J}_k \equiv \hat{J}_i \cdot (\hat{J}_j \times \hat{J}_k)$ term (Figure 13b) is essentially the quantum mechanical version of the classical geometric expression for the volume of a region spanned by three vectors $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ (Figure 13a). The factor of $1/8$ ensures that for each set of three edges only the volume of the quadrant spanned by those edges is counted.

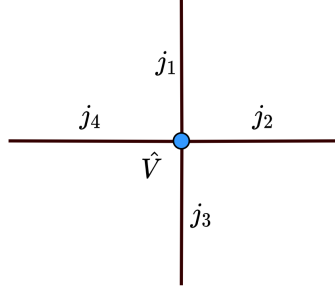


Figure 14: A 4-valent vertex with the vertex intertwiner shown as a four-index tensor

Using these expression we can now ask what is the “volume” of the quantum tetrahedron shown in Figure 10b. To make matters simple let us assume that all of the angular momentum operators are of the spin $1/2$ representation. In this case the Hilbert space on which the volume operator will act is given by the tensor product space:

$$\mathcal{H} = \mathcal{H}_{\frac{1}{2}} \otimes \mathcal{H}_{\frac{1}{2}} \otimes \mathcal{H}_{\frac{1}{2}} \otimes \mathcal{H}_{\frac{1}{2}} \quad (29)$$

The requirement of gauge invariance restricts the allowed space of states to the subspace $\text{Inv}(\mathcal{H})$ invariant under $SU(2)$ rotations at each of the vertices. From the representation theory of $SU(2)$ we know that the tensor product of four spin $1/2$ representations can be written as a direct sum of irreducible representations as follows:

$$1/2 \otimes 1/2 \otimes 1/2 \otimes 1/2 = 0 \oplus 0 \oplus 1 \oplus 1 \oplus 2 \quad (30)$$

The physical states live in the two copies of scalar (“s-wave”) states. Two non-orthogonal states which span this Hilbert space can be written as [40]:

$$|\Psi_1\rangle = \frac{1}{2} (|1001\rangle - |0101\rangle + |0110\rangle - |1010\rangle) \quad (31)$$

$$|\Psi_2\rangle = \frac{1}{2} (|1001\rangle - |0011\rangle + |0110\rangle - |1100\rangle) \quad (32)$$

We can construct two orthogonal basis states for the spin-singlet Hilbert space by orthogonalizing the above two states.

The two states which span this two dimensional invariant subspace can be written as:

$$\begin{aligned} |\Psi_1\rangle &= |\Psi_A\rangle + |\Psi_B\rangle \\ |\Psi_2\rangle &= |\Psi_A\rangle - |\Psi_B\rangle \end{aligned} \quad (33)$$

where:

$$\begin{aligned} |\Psi_A\rangle &= |0101\rangle + |1010\rangle \\ |\Psi_B\rangle &= |1001\rangle + |0110\rangle \end{aligned} \quad (34)$$

Here the $|0\rangle, |1\rangle$ are the two possible states of each edge carrying a spin 1/2 label. One can then write down the effective degrees of freedom carried by a four-valent intertwiner in terms of a single spin 1/2 state $|\tilde{\Psi}\rangle$ spanned by the invariant basis states $\{|\Psi_1\rangle, |\Psi_2\rangle\}$:

$$|\tilde{\Psi}\rangle = \alpha|\Psi_1\rangle + \beta|\Psi_2\rangle \quad (35)$$

. The states $\{|\Psi_1\rangle, |\Psi_2\rangle\}$ can also be expressed in a manner which makes the connection with tensor network states apparent. Any state of N spins can be written as:

$$|\Psi\rangle = i^{A_1, \dots, A_N} |i_1 i_2 \dots i_N\rangle \quad (36)$$

where $\{|i_k\rangle\}$; $k \in 1, \dots, D$, with D being the dimension of the site Hilbert space. Thus we can write:

$$\begin{aligned} |\Psi_1\rangle &= i_1^{ABCD} |i_A i_B i_C i_D\rangle \\ |\Psi_2\rangle &= i_2^{ABCD} |i_A i_B i_C i_D\rangle \end{aligned} \quad (37)$$

where the two 4-index tensors i_1^{ABCD} and i_2^{ABCD} live on the vertex. These states can be represented using the diagrammatic tensor network notation as shown in [Figure 15](#).

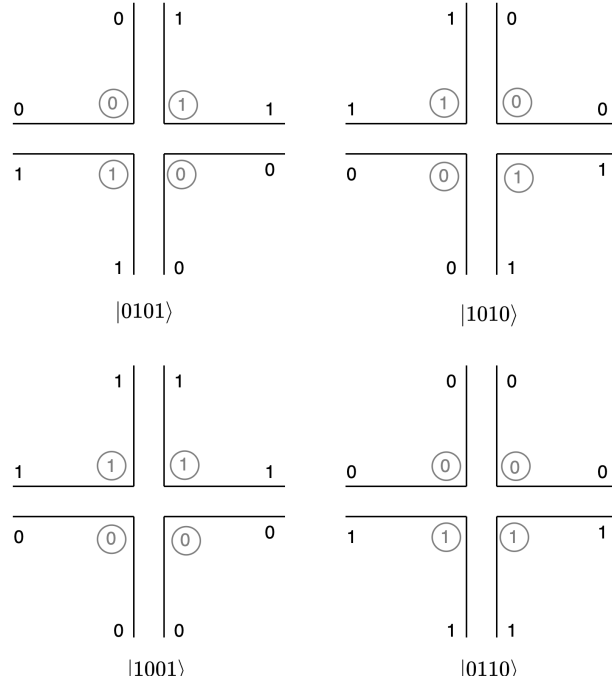


Figure 15: Tensor network representation of the 4-valent intertwiner state. The “first” qubit of each state lives on the top edge and then we proceed counter-clockwise around the vertex.

6 CZX State

In this section a model for a SPT phase with a Z_2 symmetry in a many-body system which admits a particularly clear mapping to the spin-network structure of quantum geometric states. This is the CZX model [41] (named after the unitary operators C , Z and X which are used to construct the state). Consider the system shown in Figure 16a. Here each site consists of four spins. The spins between neighboring sites are connected as shown in Figure 16a. The connected spins in four neighboring sites form a *plaquette*. The state we will construct, following [41] will be invariant under the action of Z_2 symmetry at each site.

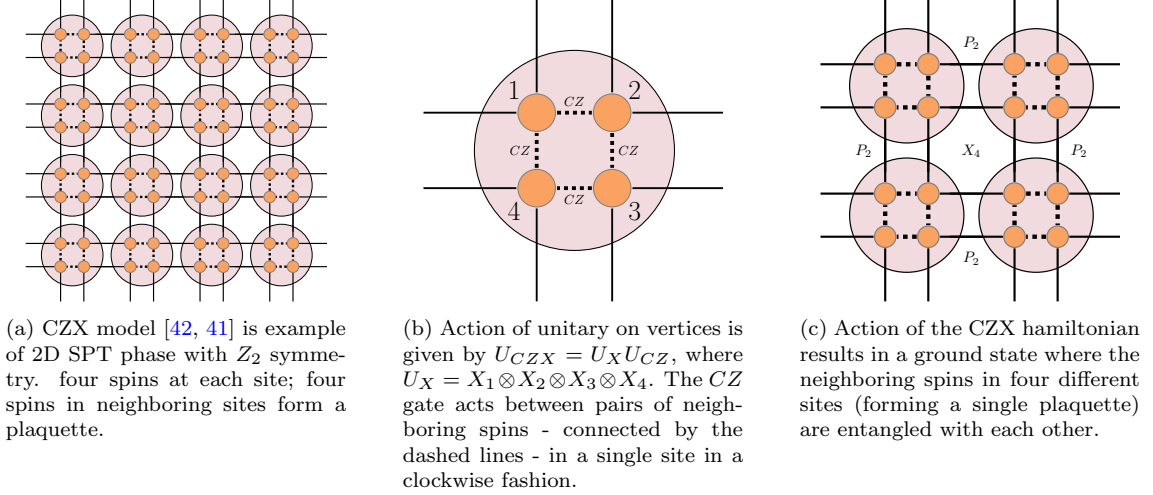


Figure 16: The CZX state with SPT and an on-site Z_2 symmetry

This state is generated by the action of two operators: U_X and U_{CZ} , which are defined as follows:

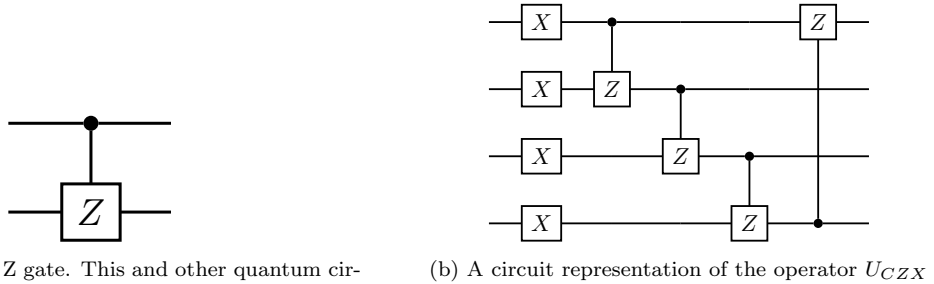
$$U_X = X_1 \otimes X_2 \otimes X_3 \otimes X_4 \quad (38a)$$

$$U_{CZ} = (CZ)_{12}(CZ)_{23}(CZ)_{34}(CZ)_{41} \quad (38b)$$

where $\{X_i\}$ are the usual Pauli spin flip matrices acting on the four spins of a given site as shown in Figure 16b. $\{CZ_{ij}\}$ is the controlled Z gate acting on two spins, which can be written in operator form as:

$$CZ = |00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10| - |11\rangle\langle 11|. \quad (39)$$

If the state of the first spin is $|0\rangle$, the state of the second spin is left unchanged. If first spin is in $|1\rangle$ state, then the (negative) Z operator is applied to the second spin. Written as a quantum circuit this operator can be also expressed in the form shown in Figure 17a. The U_X and U_{CZ} gates generate a Z_2 group. This can be seen since both square to $\mathbb{1}$ and also commute with each other.



The Hamiltonian for the whole system is given by $H = \sum H_{p_i}$, where p_i represents the i^{th}

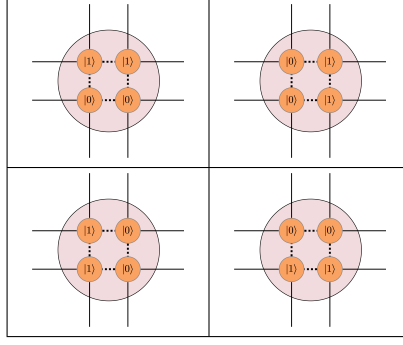


Table 3: The “code” subspace of the vertex Hilbert space which is mapped to itself under the action of U_{CZ} and U_X . E.g. $U_X|1100\rangle = |0011\rangle$ and $U_{CZ}|1100\rangle = -|1100\rangle$

plaquette. The Hamiltonian for the i^{th} plaquette is given by:

$$H_{p_i} = -\mathbf{X}_4 \otimes P_2^u \otimes P_2^d \otimes P_2^l \otimes P_2^r \quad (40)$$

Here $\mathbf{X}_4$ is the four qubit operator given by:

$$\mathbf{X}_4 = |0000\rangle\langle 1111| + |1111\rangle\langle 0000|, \quad (41)$$

which acts on the four spins in neighboring sites (forming a single plaquette) as shown in Figure 16c. The letter $\mathbf{X}$ is used for this operator because it performs the same function as the Pauli X gate, only acting on groups of four spins at a time. This operator ensures that the spins within each plaquette are all in the same state - either $|0\rangle$ or $|1\rangle$.

The P_2^i operators, with $i \in \{u, d, l, r\}$, act on the two spins in the neighboring half-plaquettes as shown in Figure 16c. These operators have the form:

$$P_2^i = |00\rangle\langle 00| + |11\rangle\langle 11|, \quad (42)$$

and ensure that each term in the Hamiltonian satisfies the on-site Z_2 symmetry. All the terms in the Hamiltonian commute with each other:

$$[H_{p_i}, H_{p_j}] = 0, \quad \forall i \neq j, \quad (43)$$

and consequently the ground state of the total Hamiltonian will simply be the product of the ground state of the i^{th} plaquette. It can easily be checked that this state is given by:

$$|\psi_{p_i}\rangle = |0000\rangle + |1111\rangle \quad (44)$$

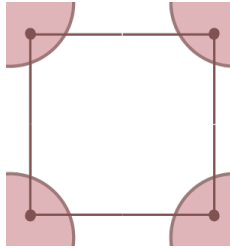


Figure 18: Each plaquette is in an entangled state: $|0000\rangle + |1111\rangle$

7 SPT Phase of Quantum Geometry

It is now straightforward to map the CZX state described in the previous section to a state of quantum geometry. This is possible because the structure of the lattice sites in the CZX state is identical to that of the Hilbert space of a single tetra-valent spin network vertex with all edges labeled by $j = 1/2$ representations of $SU(2)$. The object seen in Figure 16b can therefore be viewed as a single 4-valent vertex.

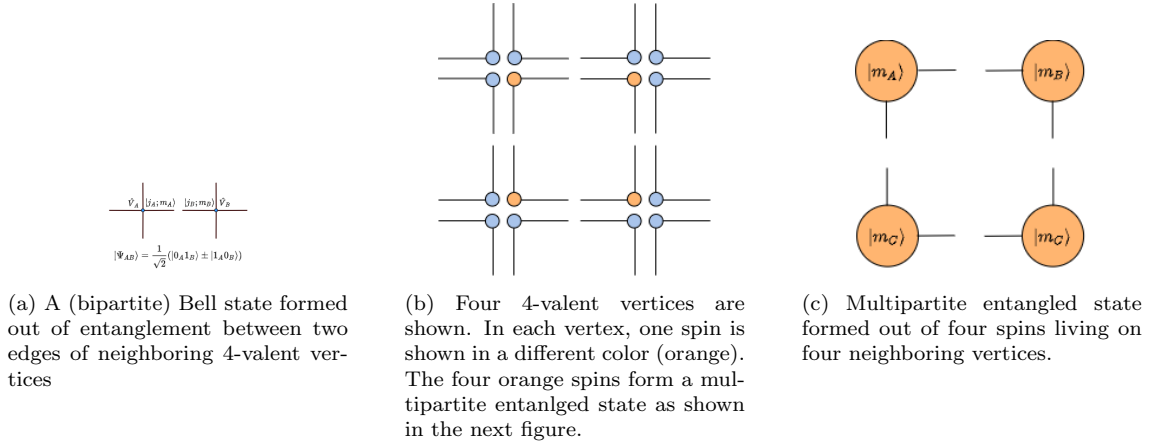


Figure 19: Spin network states with bipartite and multipartite entanglement between neighboring vertices

8 Z_2 Action and Time Reversal in Spin Networks

9 Discussion

As explained in [Appendix C](#) it is only in the tetrad (Palatini) formulation of general relativity that a change in the sign of the lapse function has any physical effects. Changing the sign of the lapse $N(t) \rightarrow -N(t)$ causes the determinant of the *tetrad* field to change sign $\det({}^4e) \rightarrow -\det({}^4e)$. Spin networks are objects which, as such, live on purely spatial hypersurfaces and have no knowledge of the lapse function or other quantities which require information about the spacetime at other time slices. Therefore one might ask whether it makes sense to speak about time reversal at the level of spin networks at all and even if it does, how can one possibly connect something which requires knowledge of time evolution (change of the sign of the lapse) to something which does not possess such knowledge (spin networks encoding the geometry of spatial hypersurfaces)? we will address both of the concerns in turn.

Insert figure of ADM split with spin networks

9.1 Time Reversal in Spin Networks

The first criticism is the following. Spin networks are purely spatial objects. The “time evolution” of spin networks is generated by the Hamiltonian constraint operator of LQG and results in four-dimensional simplicial complexes known as “spin foams”. If one wishes to talk about time reversal then would not spin foams, rather than spin networks, be the appropriate arena?

It is true that a more complete discussion of time reversal in LQG should take place at the level of spin foams. However, firstly, such an analysis would be significantly more involved than the discussion presented in this work. Second, consider a simple system such as a single spin $|\uparrow\rangle$ particle moving in ordinary flat space. Applying time reversal to this system simply flips the spin of the particle $|\downarrow\rangle$. One can do so without having any knowledge of the past or future trajectory of the particle in question. Whether or not the entire worldline of the particle is symmetric under time reversal is a separate question and the possible types of trajectories (symmetric or anti-symmetric under time reversal) are used to classify different types of order in condensed matter systems such as in spin hall insulators.

In the same vein, one can always write down the results of applying time-reversal to any spin-network. One simply has to flip all the spins labeling the edges or, if one wishes, flip only the spins in a given finite subset of the graph network where we want time-reversal to act. Whether the spin-foam, resulting from the time evolution of the time reversed spin network state, is time symmetric or not is a separate question⁹. Thus, examining the properties of spin networks under the action of time-reversal is a completely valid line of investigation.

⁹However, one would expect that spin-networks which violate time-reversal symmetry would also generate spin-foam histories which are not time-reversal invariant.

9.2 Time Reversal, Spatial Volume and the Lapse

The second criticism concerns being able to connect the sign of the lapse to spin networks. The lapse is a quantity which tells us how spatial slices at different times evolve into each other. Spin networks on the other hand are purely spatial objects with knowledge only of the spatial geometry in a given time slice. Why should one know about the other? It turns out that this criticism already has its resolution built into the structure of spin networks.

When we apply time-reversal to a portion of the spin network we flip the spins living on the edges which fall in that given region. Doing so is equivalent to changing the sign of the triad field: $e_a^i \rightarrow -e_a^i$, because as explained in the discussion in the text centered around (D.21), the quantum operator corresponding to the triad field is the $SU(2)$ angular momentum operator $\hat{J}_i$.

Now if we change the sign of the triad field, this results in changing the sign of the determinant $\det(^3e)$ of the triad field as well¹⁰. This implies that the volume of the classical geometry described by the given spin network also changes sign:

$$\text{Vol} = \int d^3x \det(^3e) \rightarrow -\text{Vol} \quad (45)$$

Note, that this only happens for the triad and not for the tetrad! Therefore if we implement time-reversal on a spin-network in the manner described in this work the resulting *four-dimensional* volume element will also undergo a change in sign $\det(^4e) \rightarrow -\det(^4e)$. Since the determinant of the tetrad and triad fields are related to each other¹¹ by a factor of the lapse function $\det(^4e) = N(t) \det(^3e)$, changing the sign of the determinant of the triad field is *equivalent* to changing the sign of the lapse function.

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¹⁰For an $N \times N$ matrix $\mathbf{A}$, changing the matrix elements $\mathbf{A} \rightarrow -\mathbf{A}$ the change in the sign of the determinant is given by $\det(\mathbf{A}) = (-1)^N \det(-\mathbf{A})$

¹¹This follows from the form (C.8) of the 4-metric $g_{\mu\nu}$ written in terms of the lapse, shift and the 3-metric h_{ab} , from which one can deduce that $\sqrt{\det(g)} = N \sqrt{\det(h)}$.

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A Time Reversal of Quantum Systems

In classical mechanics the action of time-reversal symmetry on physical observables can be defined as shown in Figure 20

Under time-reversal the Poisson brackets change sign:

$$\{x_i, p_j\} = \delta_{ij} \rightarrow \{x'_i, p'_j\} = -\delta_{ij} \quad (\text{A.1})$$

where $\mathbf{x}' = \mathcal{T}\mathbf{x} = \mathbf{x}$, $\mathbf{p}' = \mathcal{T}\mathbf{p} = -\mathbf{p}$ and $\mathcal{T}$ is the generator of time-reversal. However, the classical equations of motion remain invariant:

$$\dot{\mathbf{x}}' = \frac{\partial H}{\partial \mathbf{p}'}; \quad \dot{\mathbf{p}}' = -\frac{\partial H}{\partial \mathbf{x}'} \quad (\text{A.2})$$

In quantum mechanics the situation is quite different and this difference can ultimately be traced back to the appearance of the imaginary $i = \sqrt{-1}$ in the canonical commutation relations and the Schrodinger equation. If, proceeding by analogy with the classical case, we implement time reversal for quantum operators in the expected way:

$$\hat{\mathbf{x}} \rightarrow \hat{\mathbf{x}}; \quad \hat{\mathbf{p}} \rightarrow -\hat{\mathbf{p}} \quad (\text{A.3})$$

the quantum commutators pick up a negative sign, similar to what happens in the classical case (A.1):

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij} \rightarrow [\hat{x}'_i, \hat{p}'_j] = -i\hbar\delta_{ij} \quad (\text{A.4})$$

Physical Observable	Under time reversal
Position	$\mathbf{x} \rightarrow \mathbf{x}$
Time	$t \rightarrow -t$
Linear Momentum	$\mathbf{p} \rightarrow -\mathbf{p} = d\mathbf{x}/d(-t) = -d\mathbf{x}/dt$
Angular Momentum	$\mathbf{l} = \mathbf{x} \times \mathbf{p} \rightarrow -\mathbf{l}$
Energy/Hamiltonian	$H(\mathbf{x}, \mathbf{p}) \rightarrow H(\mathbf{x}, -\mathbf{p})$
Electric Field	$E^i \rightarrow -E^i$
Magnetic Field	$B^i \rightarrow -B^i$

Figure 20: Action of time-reversal symmetry on classical observables

However, the equation of motion is no longer invariant:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi \rightarrow i\hbar \frac{\partial \psi}{\partial t'} = -\hat{H}\psi \quad (\text{A.5})$$

where $t' = -t$. This can be resolved, however, if following Wigner we define the quantum time reversal operator $\mathcal{T}$ to act as follows on any quantum state as follows:

$$\mathcal{T}(c_1\psi_1 + c_2\psi_2) = c_1^*\mathcal{T}\psi_1 + c_2^*\mathcal{T}\psi_2 \quad (\text{A.6})$$

Such an operator is called *anti-linear* or *anti-unitary*. Acting with this operator on both sides of the Schrodinger equation we obtain:

$$\mathcal{T} \left(i\hbar \frac{\partial \psi}{\partial t} \right) = i\hbar \frac{\partial (\mathcal{T}\psi)}{\partial t'} = H(\mathcal{T}\psi) \quad (\text{A.7})$$

where in the second step we have used the fact that $i \rightarrow -i$ under the action of $\mathcal{T}$ and that $t' = -t$. Consequently, if ψ satisfies the Schrodinger equation, then $\mathcal{T}\psi$ satisfies the time-reversed Schrodinger equation. A second benefit of this definition is that under $\mathcal{T}$, the quantum commutators no longer change sign:

$$\mathcal{T}([\hat{x}_i, \hat{p}_j])\mathcal{T}^{-1} = \mathcal{T}i\mathcal{T}^{-1}\hbar\delta_{ij} \rightarrow [\hat{x}'_i, \hat{p}'_j] = i\hbar\delta_{ij} \quad (\text{A.8})$$

where we have used the fact that an operator $\mathcal{O}$ transforms as $U\mathcal{O}U^{-1}$ under the action of any symmetry U . As a sidenote let us mention that while the Wigner definition of time-reversal leads to an invariant equation of motion *and* invariant commutators in the quantum case, the definition of time-reversal in the classical case leaves the equation of motion invariant, but not the Poisson brackets. This dichotomy between the classical and quantum cases deserves further study, but we postpone that for a later occasion.

B ADM Splitting and Tetrads

Recall that in the ADM formulation the four-dimensional spacetime metric can be written in terms of a lapse function $N(x^\mu)$ and a spatial shift vector field $N^i(\vec{x})$:

$${}^4g_{\mu\nu} = (-N^2 + N^i N_i)dt^2 + 2N_i dt dx^i + h_{ij} dx^i dx^j \quad (\text{B.1})$$

where $i \in \{1, 2, 3\}$, $\mu, \nu \in \{0, 1, 2, 3\}$, $N(x^\mu)$ is the lapse function, $N^i(\vec{x})$ is the shift vector field with $\vec{x} \equiv (x, y, z)$ being a purely spatial co-ordinate; and h_{ij} is the internal metric of the constant time surfaces. This metric can be expressed in terms of a tetrad (“vielbein” or local Lorentz frame) as:

$$g_{\mu\nu} = e_\mu^W e_\nu^J \eta_{IJ}$$

where $\eta_{IJ} = \text{diag}(-1, 1, 1, 1)$ is the flat Minkowski metric. In the so-called time gauge the time-time component of the above metric (B.1) can be used to determine the tetrad components for $e_0^W e$:

$$e_0^0 = N(x^\mu)dt; \quad e_i^0 = N^i(\vec{x})dx_i \quad (\text{B.2})$$

with $i \in \{1, 2, 3\}$, and all other components of the tetrad set to zero. There is no summation over repeated indices in this expression. The remaining components of the tetrad field require knowledge of the internal metric and also a choice of local Lorentz frame. We can always choose a set of coordinates, locally, in which the spatial metric becomes diagonal:

$$h_{ij} \equiv \text{diag}(h_1, h_2, h_3) \quad (\text{B.3})$$

The tetrad components for $e_i^W e$, $i \in \{1, 2, 3\}$ then can be chosen to be:

$$e_i^i = \sqrt{h_i}, \quad (\text{B.4})$$

with the rest being set to zero. When the spatial metric is flat and the shift vector is set to zero, the only non-zero components of the tetrad field will be given by:

$$e_0^0 = N(t)dt; \quad e_i^0 = dx^i$$

C Time Reversal in Classical GR

In [23, 24] the question of discrete symmetries in LQG is considered and it is shown that in the presence of fermions the sign of the lapse function can, in principle, be detected. This can be seen

as follows. Consider a geometry where the shift vector is identically zero, the lapse vector is a function only of the time coordinate and the spatial metric is the flat Euclidean metric. In this case the tetrad components become ¹²:

$$e_0^0 = N(t)dt; \quad e_i^0 = dx^i, \quad (C.1)$$

Now, when fermions are present both the tetrad field and the gravitational spin connection ω_{μ}^{IJ} play a role in determining their dynamics via the Dirac equation:

$$\gamma_W e e_{\mu}^W e \left[\partial^{\mu} + \frac{i}{2} \omega_{KL}^{\mu} \tau^{KL} \right] \psi + m \mathbb{1} \psi = 0, \quad (C.2)$$

where $\{\gamma_I\}$ are the Dirac matrices, τ^{KL} are the generators of the Lorentz group and $\mathbb{1}$ is the identity matrix included here for clarity. The spin connection for the given geometry (C.1) vanishes. Assuming that ψ is a function only of time, the above expression reduces to:

$$\gamma_0 N(t) \partial_t \psi + m \mathbb{1} \psi = 0. \quad (C.3)$$

This expression has a straightforward solution given by:

$$\psi(t) = e^{imf(t)} \psi_0 \quad (C.4)$$

where the phase $f(t)$ is given by:

$$f(t) = \int_{t_0}^t N(t) dt \quad (C.5)$$

Since the phase in (C.4) is a function of the lapse one can, in principle, design an interference experiment involving two fermions with different masses whose result would depend on the sign of the lapse function. Alternatively one can describe the behavior of the fermion field in a region of negative lapse as evolving “backwards in time”. Since, in principle, nothing forbids the lapse from taking on negative values this leads us to the possibility of the existence of “pockets” of geometry where time “runs backwards” as compared to the surrounding regions.

In terms of the action principle, the ambiguity in the sign of the lapse can be seen as arising due to an ambiguity in the determinant of the tetrad matrix. The usual Einstein-Hilbert action in the metric formalism is:

$$S_{EH} = \frac{1}{\kappa} \int d^4x \sqrt{-g} R + \int d^4x \sqrt{-g} \mathcal{L}_{matter}, \quad (C.6)$$

where $\kappa = 8\pi G_N$, R is the Ricci scalar and $g = \det(g)$ is the determinant of the 4-metric. Expressed in terms of the tetrad and connection variables [22, Sec 4.5.3] the action becomes:

$$\begin{aligned} S_P[e, \omega] &= \frac{1}{2\kappa} \int d^4x \star (e^I \wedge e^J) \wedge F^{KL} \epsilon_{Wew eKL} \\ &= \frac{1}{4\kappa} \int d^4x \epsilon^{\mu\nu\alpha\beta} \epsilon_{W eJKL} e_{\mu}^I e_{\nu}^J F_{\alpha\beta}^{KL} \\ &= \frac{1}{4\kappa} \int d^4x (\det^4 e) R[e] \end{aligned} \quad (C.7)$$

where $F_{\alpha\beta}^{KL}$ is the curvature of the spin-connection ω_{μ}^{IJ} and $\det^4(e)$ is the determinant of the *tetrad* field. From the form of this action and the relation (C.1), which implies that $\det e \propto \text{sign}(N(t))$, we can see that the sign of the tetrad action S_P depends on the sign of the lapse. The sign of the lapse, however, has no effect on the sign of the determinant of the 4-metric and therefore does not affect the action given in (C.6). One can see this more concretely by noting that in terms of the ADM lapse and shift fields, the 4-metric can be expressed as follows [22, Eq (4.14)]:

$$g_{\mu\nu} = \begin{pmatrix} -N^2 + N^a N_a & \mathbf{N} \\ \mathbf{N}^T & h_{ab} \end{pmatrix} \quad (C.8)$$

¹²For a reminder of how the lapse function can be related to tetrad components we refer the reader to [Appendix B](#)

where $\mathbf{N}$ is the spatial shift vector field, $N(t)$ is the lapse function and h_{ab} is the metric on the spatial hypersurfaces. Since only square of the lapse occurs in the 4-metric, the sign of the lapse does not affect its determinant $\sqrt{-\det g}$. Therefore, it is only in the tetrad-connection formalism that a *local* change in the sense of time-evolution and represented by a change in the sign of the lapse can have an effect on the dynamics.

D Quantum States of Geometry

The key point of this paper is that time-reversal symmetry can be made into a local gauge field acting on quantum states of spacetime geometry. These states of quantum geometry, known as “spin networks”, originate in an approach towards a theory of quantum gravity known as *loop quantum gravity* [44, 22]. Here we outline the key steps leading from classical gravity to its quantization in terms of spin-networks. The notation rules are as follows: lower case greek letters are 3 + 1 dimensional spacetime indices $\alpha, \beta, \dots \in \{0, 1, 2, 3\}$, upper case latin letters are $\mathfrak{so}(3, 1)$ Lie-algebra indices $I, J, \dots \in \{0, 1, 2, 3\}$, lower case latin letters from the beginning of the alphabet are three-dimensional spatial indices $a, b, c \in \{1, 2, 3\}$ and those from the middle of the alphabet are $\mathfrak{su}(2)$ Lie algebra valued indices $i, j, k \in \{1, 2, 3\}$.

We will divide the process into several parts: the first part describes the path from the usual metric theory of gravity to the connection formalism and the form of the ADM constraints; the second describes the quantization strategy and how we obtain graph states as the candidate quantum states of geometry; the third explains how to impose the Gauss constraint to obtain gauge invariant states of quantum geometry; and the fourth elaborates on how the intertwiner (or vertex) degrees of freedom come about from the imposition of gauge invariance.

A. From metric to connection variables:

A.1 Starting with the usual Einstein-Hilbert action in the 1st order formalism:

$$S_{EH} = \frac{1}{\kappa} \int d^4x \sqrt{-g} R[g, \Gamma], \quad (\text{D.1})$$

where $\kappa = 8\pi G_N$, R is the Ricci scalar and $g = \det(g)$ is the determinant of the 4-metric, we perform a canonical transformation [45, 46] from the four-metric and Christoffel connection being our dynamical variables $\{g_{\mu\nu}, \Gamma_{\alpha\beta}^\delta\}$ to a new set of variables $\{e_\mu^I, A_\mu^{IJ}\}$ (known as the Ashtekar or Ashtekar-Barbero variables), where $e_\mu^W e$ is a tetrad field (choice of local Lorentz frame) and A_μ^{IJ} is a $\mathfrak{so}(3, 1)$ Lorentz gauge connection.

A.2 In the new variables the Einstein-Hilbert action can be written in the form:

$$\begin{aligned} S_P[e, A] &= \frac{1}{2\kappa} \int d^4x \star (e^I \wedge e^J) \wedge F^{KL} \epsilon_{W e JKL} \\ &= \frac{1}{4\kappa} \int d^4x \epsilon^{\mu\nu\alpha\beta} \epsilon_{W e JKL} e_\mu^I e_\nu^J F_{\alpha\beta}^{KL} \end{aligned} \quad (\text{D.2})$$

where $F_{\alpha\beta}^{KL}$ is the curvature of the spin-connection A_μ^{IJ} .

A.3 Next we perform a 3 + 1 split of spacetime following the usual ADM (Arnowitt-Deser-Misner) prescription which yields the following expression for the Einstein-Hilbert-Palatini action:

$$S_P = \int dt \int_M d^3x \left(E_{IJ}^\mu \dot{A}_\mu^{IJ} - N\mathcal{H} + N_a \mathcal{C}^a + \Lambda_{IJ} \mathcal{G}^{IJ} \right) \quad (\text{D.3})$$

where A_μ^{IJ} is the configuration variable, $\dot{A}_\mu^{IJ}$ is the generalized “velocity” and:

$$E_{IJ}^\mu = \frac{1}{2} \epsilon_{IJKL} \epsilon^{abc} e_a^K e_b^L \quad (\text{D.4})$$

is the momentum conjugate to A ; ϵ^{abc} is the anti-symmetric tensor living only on the spatial slices. The inner integral is over the spatial 3-manifolds M which are the “leaves” of our 3 + 1 decomposition.

A.4 One can now perform a Legendre transformation to obtain the Hamiltonian for general relativity in connection variables:

$$H_{LQG} = N\mathcal{H} - N_a\mathcal{C}^a - \Lambda_{IJ}\mathcal{G}^{IJ} \quad (\text{D.5})$$

Note that, since N , N^a and Λ_i in (D.3) are all Lagrange multipliers, the associated terms $\mathcal{H}$, $\mathcal{C}^a$ and $\mathcal{G}^i$ are all constraints, *i.e.* they are functions which have to vanish on the space of solutions. N is known as “lapse” function, N^a is the “shift” vector field. In contrast to the usual ADM prescription, in the connection formulation we get an additional constraint $\Lambda_{IJ}\mathcal{G}^{IJ}$, where $\mathcal{G}^{IJ}$ has the form:

$$\mathcal{G}_{IJ} = \mathcal{D}_a E_{IJ}^a \equiv 0 \quad (\text{D.6})$$

and is known as the “Gauss” constraint.

A.5 Here the classical variables are a Lorentz $\mathfrak{so}(3,1)$ connection ω_μ^{IJ} and an “electric field” E_{IJ}^μ of the Yang-Mills type. When restricted to the spatial manifold these variables turn into a $\mathfrak{su}(2)$ connection A_a^i and a densitized triad field $\tilde{E}_i^a = \sqrt{\det(h)}e_i^a$, where h_{ab} is the 3-metric on M and e_i^a is an orthonormal vector field spanning the local tangent space $T_p(M)$ at any point of M . The Gauss constraint (D.6) can then be written in the form:

$$\mathcal{G}_i = \mathcal{D}_a \tilde{E}_i^a = 0 \quad (\text{D.7a})$$

$$\Rightarrow \oint_{S^2} E_i^a n_a d^2x = 0 \quad (\text{D.7b})$$

where in the second line we have integrated the differential form of the Gauss constraint over a closed two-dimensional surface of spherical topology with n_a being the normal to the area element at any point on the surface. This, then, becomes the form of “Gauss’ Law” which says that - in the absence of any charged matter - the total electric flux through any closed surface is zero. We shall see that this statement is crucial for constructing the physical states of quantum geometry.

B. From connection gravity to graph states

B.1 Quantization proceeds by using the intuition gained from quantizing gauge theories of the Yang-Mills types. In Yang-Mills theories, rather constructing operators living in $L^2(M)$ (where M is the spatial manifold on which our quantum fields live) one constructs gauge-invariant observables of the Wilson type and promotes these to operators in the quantum theory. For the connection variable this means using as the classical variable the holonomies along a Wilson loop:

$$h(A) = \int_\gamma dx^\mu A_\mu^I(x) \tau_{We} \quad (\text{D.8})$$

B.2 The key difference between loop quantum gravity and other theories is that in LQG the background spacetime is *itself* a quantum entity and therefore while performing the quantization we do not assume the presence of a pre-existing background geometry. Thus, while constructing the quantum phase space we should be careful to ensure that we work only with background independent operators. We can do so by [47, Sec 3.2] using holonomies to probe the background geometry, however, since we need to probe the geometry over an extended region and not only along a single curve, instead of individual curves we use arbitrary graphs with a holonomy $h_e(A)$ attached to each edge. The quantum configuration space associated with a given graph Γ , with L edges becomes:

$$\Psi_\Gamma(A) = \psi(h_{e_1}(A), \dots, h_{e_L}(A)) \quad (\text{D.9})$$

where ψ is function on L copies of the gauge group G of the theory which in our case happens to be $SU(2)$.

B.3 Such functions are known as “cylindrical” functions. We will write the space of cylindrical functions *w.r.t.* a graph Γ as Cyl_Γ . An inner-product on Cyl_Γ can be given using the Haar measure on $SU(2)$:

$$\langle \Psi_1 | \Psi_2 \rangle_\Gamma := \int_{G^L} d\mu_L \bar{\psi}_1 \psi_2 \quad (\text{D.10})$$

B.4 Now, states defined on a single graph will naturally represent only a small sector of all possible states on the space of connections $\mathcal{A}$. As shown in [48] the above measure can be extended to a space of “generalized connections” $\bar{\mathcal{A}}$ which consists of states living on *arbitrary* graphs.

B.5 Using the inner product (D.10) the Hilbert space for a single graph Γ becomes:

$$\mathcal{H}_\Gamma = L^2(SU(2)^L), \quad (\text{D.11})$$

i.e. the space of square integrable functions living on L copies of $SU(2)$.

B.6 we can use the Peter-Weyl theorem to perform a sort of Fourier transform on states living in $\mathcal{H}_\Gamma$ and write them in terms of spin-labels $\{j_{e_i}\}$ associated with each edge rather than with function on the group manifold:

Peter-Weyl Theorem. *The matrix elements of the Wigner representation matrices of a compact Lie group are orthogonal w.r.t. the Haar measure on the group:*

$$\int d\mu(g) \overline{D_{m'n'}^{j'}(g)} D_{mn}^j(g) = \frac{1}{d_j} \delta^{jj'} \delta_{mm'} \delta_{nn'}, \quad (\text{D.12})$$

where $d\mu(g)$ is the Haar measure, $d_j = (2j+1)$ is the dimension of the j^{th} representation of $SU(2)$ and $D_{mn}^j(g)$ representation matrix of the element $g \in G$. This theorem implies that any square integrable function $f(g)$ on G can be expressed as a linear sum over the group representation matrices:

$$f(g) = \sum_{j;mn} f_j^{mn} D_{mn}^j(g) \quad (\text{D.13})$$

Each representation matrix is $(2j+1) \times (2j+1)$ dimensional and can therefore be thought of [44, Sec 5.2] as a map from the Hilbert space $\mathcal{H}_j$ of a spin j particle to itself:

$$D^j : \mathcal{H}_j \rightarrow \mathcal{H}_j \quad (\text{D.14})$$

Each such matrix can therefore also be viewed as an element of $\mathcal{H}_j \otimes \mathcal{H}_j$. Consequently the space $L^2(G)$ can be decomposed as a direct sum over all possible representations of $SU(2)$:

$$L^2(G) = \oplus_j (\mathcal{H}_j \otimes \mathcal{H}_j) \quad (\text{D.15})$$

Note: this implies that with each edge there are associated *two* copies of the Hilbert space of a spin j particle. This fact will be crucial going forwards.

B.7 Each graph Γ has L edges with a total Hilbert space given by (D.11). Using (D.15) this space can then be written as a tensor product over the the Hilbert space of each edge:

$$\mathcal{H}_\Gamma = L^2(SU(2)^N) = \otimes_e [\oplus_{j_e} (\mathcal{H}_{j_e} \otimes \mathcal{H}_{j_e})] \quad (\text{D.16})$$

Let $e_{vv'}$ be the (directed) edge connecting the two vertices labeled v and v' with v being the source of the edge and v' the target. We associate one of the two Hilbert spaces living on $e_{vv'}$ with v and the other with v' . The Hilbert space of a given edge can thus be written as:

$$\mathcal{H}_e = \oplus_{j_e} (\mathcal{H}_{j_e}^{e,v} \otimes \mathcal{H}_{j_e}^{e,v'}), \quad (\text{D.17})$$

keeping in mind that each j is associated with a particular edge. Equivalently we can label the source vertex of the edge e as $v \equiv v_{e+}$ and the target vertex as $v' \equiv v_{e-}$. Then the above equation becomes:

$$\mathcal{H}_e = \oplus_{j_e} (\mathcal{H}_{j_e}^{v_{e+}} \otimes \mathcal{H}_{j_e}^{v_{e-}}) \quad (\text{D.18})$$

B.8 Using this notation we can rewrite the Peter-Weyl decomposition a little differently by grouping together the Hilbert spaces associated with each vertex rather than the Hilbert spaces associated with each edge. The Hilbert space associated with a single *vertex*, becomes:

$$\mathcal{H}_v = \bigoplus_{\{j_e\}} \left[\bigotimes_{e^+ \in E_v^+} \mathcal{H}_{j_e}^{v_{e^+}} \bigotimes_{e^- \in E_v^-} \mathcal{H}_{j_e}^{v_{e^-}} \right], \quad (\text{D.19})$$

where E_v^+, E_v^- are the sets of all outgoing and ingoing edges, respectively, connected to v and the outermost tensor sum is over the set of all possible labeling of edges in $E_v = E_v^+ \cup E_v^-$ with representations of $SU(2)$.

B.9 The total Hilbert space of the graph Γ then becomes:

$$\mathcal{H}_\Gamma = \bigotimes_{v \in \Gamma} \mathcal{H}_v = \bigotimes_{v \in \Gamma} \left\{ \bigoplus_{\{j_e\}} \left[\bigotimes_{e^+ \in E_v^+} \mathcal{H}_{j_e}^{v_{e^+}} \bigotimes_{e^- \in E_v^-} \mathcal{H}_{j_e}^{v_{e^-}} \right] \right\} \quad (\text{D.20})$$

C. From graph states to spin networks

C.1 The Hilbert space obtained in (D.20) is as yet only the *kinematical* Hilbert space. This is because we have not yet imposed the three constraints $\mathcal{H}$, $\mathcal{C}^a$ and $\mathcal{G}^i$ from (D.5) which encode the physical content of general relativity. Of these three constraints $\mathcal{H}$ (not to be confused with a Hilbert space also labeled as $\mathcal{H}$) is known as the Hamiltonian or “scalar” constraint, $\mathcal{C}^a$ is called the diffeomorphism or “vector” constraint and, as mentioned previously, $\mathcal{G}^i$ is the Gauss constraint.

C.2 The Hamiltonian constraint is so named because it generates time evolution (or time-like diffeomorphisms) in the classical theory. For the time-being we will not be concerned with the application of this constraint. The graph states described above have been obtained by explicitly gauge fixing a choice of spatial foliation of the $3+1$ dimensional spacetime, and describe the geometry of a spatial slice.

C.3 The diffeomorphism constraint is so named because it generates diffeomorphisms along the spatial slices. Conveniently enough, by expressing our Hilbert space in the form (D.20) we have already imposed this constraint! This is because states in $\mathcal{H}_\Gamma$ depend only on the connectivity of the graph and the labeling of its edges by spin labels. Performing a diffeomorphism can only shift the edges or the locations of the nodes but will not alter either the graph structure or the spin labels. Thus these states are implicitly diffeomorphism invariant. In fact this was the whole motivation of [48] for constructing these graph states in the first place!

C.4 Finally we come to the Gauss constraint. In the quantum theory the configuration space is over the space of graphs labeled by holonomies of the generalized connection $\bar{\mathcal{A}}$. In describing the structure of this space we have ignored the electric field E_i^a and its operator equivalent in the quantum theory. The quantum operator corresponding to the electric field turns out to nothing more than the angular momentum operator of the group $SU(2)$ acting on the edges of the graph states¹³.

$$n_a(e) \hat{E}_i^a = \mathbf{J}_i^e, \quad (\text{D.21})$$

where $n_a(e)$ is the tangent vector to the edge e . For the special case of $j = 1/2$, these are simply the Pauli matrices $\hat{J}_i = \{\sigma_x, \sigma_y, \sigma_z\}$.

C.5 The expression (D.7b) can now be expressed in a particularly simple form when we take the surface of integration to enclose precisely one vertex v of our graph state Γ . In this event the operator version of (D.7b) can be written as:

$$\sum_{e \in E_v} \mathbf{J}_i^e = 0 \quad (\text{D.22})$$

¹³For a simple and transparent explanation of how this comes about we point the reader to [22, Sec 6.2].

Succintly, this is nothing more than the statement that *the total angular momentum carried by edges into any vertex must vanish in the absence of matter*. Somewhat less transparent is the statement that physical states must live in the $SU(2)$ invariant subspace of each vertex $\mathcal{H}_v$:

$$\mathcal{H}_\Gamma^{phys} = L^2 [SU(2)^L / SU(2)^N] = \bigotimes_{v \in \Gamma} \mathcal{H}_v^{Inv}, \quad (\text{D.23})$$

where L, N are the number of edges and the number of nodes in the graph, respectively.

C.6 States in $\mathcal{H}_v^{Inv}$ will therefore satisfy:

$$\sum_{e \in E_v} \mathbf{J}_e^\epsilon |\Psi_\Gamma\rangle = 0, \quad \forall v \in \Gamma \quad (\text{D.24})$$

D. Intertwiner Degrees of Freedom

D.1 If the edge degrees of freedom were the end of story when it comes to describing the kinematical properties of spin-networks, then we could use the result from the Peter-Weyl theorem to write down the final form of the quantum states of geometry given by (D.9) in terms of spin labels as follows:

$$\Psi_\Gamma(A) = \sum_{\{j_i\}, \{m_i\}, \{n_i\}} C_{j_1, \dots, j_L}^{m_1, \dots, m_L, n_1, \dots, n_L} D_{m_1 n_1}^{j_1} \dots D_{m_L n_L}^{j_L}, \quad (\text{D.25})$$

where the sum is over all possible representations (or “spins”) of $SU(2)$ and for each representation j_i the Wigner matrix indices take values in $m, n \in \{1, 2, \dots, 2j_i + 1\}$. However, such a state doesn’t truly capture the primary physical characteristic of a graph - the connectivity between its various edges and nodes. As such, this state only describes a graph consisting of a collection of L disconnected edges - a rather trivial structure which could not possibly capture the geometry of any region of spacetime.

D.2 What is missing, of course, is an understanding of how the *nodes* of the graph come into play. We must find some way of *sewing* all the edges together to form a graph of the form shown in Figure 10a. It turns that *this* is precisely the role the Gauss constraint plays. The constraint (D.7b) ensures that the spins of the edges coming into a single vertex v are not independent and that their representations are required to satisfy the relation (D.24). This constraint implies that the form of the coefficients of the *l.h.s.* in (D.25) cannot be arbitrary but must be such that $C_{j_1, \dots, j_L}^{m_1, \dots, m_L, n_1, \dots, n_L}$ remains invariant when any $SU(2)$ transformation acts on the set of edges coming into any given vertex.

D.3 Now, recall that the Hilbert space associated with an edge labeled by the $SU(2)$ representation j_e consists of two copies of $\mathcal{H}_{j_e}$ (c.f. (D.17) and (D.18)). Stated differently, the Wigner representation matrices for a given edge $e_{vv'}$ can be labeled as:

$$D_{mn}^{j_e} \equiv D_{m_v n_{v'}}^{j_e} \quad (\text{D.26})$$

where the first matrix index m is associated with the source vertex v and the second index n is associated with the target vertex n . Using this notation we can write the state associated with a *single* node v joined to four edges as:

$$\Psi_v(A) = \sum_{\{j_i\}, \{m_i\}} C_{j_1, \dots, j_4}^{m_1, \dots, m_4} D_{m_1 n_1}^{j_1} \dots D_{m_4 n_4}^{j_4} \quad (\text{D.27})$$

where the indices m_1, m_2, m_3, m_4 are those ends of the edges which meet the vertex and n_1, n_2, n_3, n_4 are for the “free” ends.

D.4 Now, the requirement that the coefficients $C_{j_1, \dots, j_4}^{m_1, \dots, m_4}$ remain invariant under any $SU(2)$ transformation applied to the given vertex implies that the state for this vertex must be written as:

$$\Psi_v(A) = \sum_{\{j_i\}, \{m_i\}} C_{j_1, \dots, j_4}^{m_1, \dots, m_4} D_{m_1 n_1}^{j_1} \dots D_{m_4 n_4}^{j_4} \quad (\text{D.28})$$

where $\iota^{m_1, \dots, m_4}$ is an *invariant* tensor. Applying this to the an arbitrary graph we obtain the final gauge invariant form of the spin-network wavefunctions:

$$|\Psi\rangle = \sum_{\{i_v\}, \{j_e\}} \text{Tr} \left[\prod_{v \in \Gamma} A_{i_v}^v \prod_{e \in \Gamma} D_{j_e}^e \right] |i_1, \dots, i_{n_v}; j_1, \dots, j_{n_e}\rangle \quad (\text{D.29})$$

with the tensors defined as in Table 1. We can easily see that this expression has exactly the same form as that for a tensor network with tensors $A_{i_v}^v$ describing the vertex degrees of freedom (the so-called “physical” indices) and the matrices $D_{j_e}^e$ describing the edge degrees of freedom (the “bond” indices).

E Invariant Tensors and Intertwiners

Given n particles transformation under difference representations of $SU(2)$, *i.e.* labeled by different spins $\{j_1, \dots, j_n\}$, an “invariant tensor” is a state in the tensor product state space: $\mathcal{H}^n = \bigotimes_{i=1..n} \mathcal{H}^i$; where $\mathcal{H}^i$ is the $2(j_i + 1)$ dimensional space for a spin j_i particle. States in $\mathcal{H}^n$ can be written in terms of the canonical basis:

$$|\Psi\rangle = \sum_{\{i_1, \dots, i_n\}} C_{i_1, \dots, i_n} |i_1, \dots, i_n\rangle \quad (\text{E.1})$$

where $|i_1, \dots, j_n\rangle = |i_1\rangle \otimes \dots \otimes |i_n\rangle$, with $|i_k\rangle \in \{|-j_k\rangle, |-j_k + 1\rangle, \dots, |j_k - 1\rangle, |j_k\rangle\}$ and element of the usual angular momentum eigenstates of a spin j_k system.

An invariant tensor is then any state $|\Psi_{inv}\rangle \in \mathcal{H}^n$ which satisfies the condition [49, App. A] that:

$$\sum_{i=1..n} \mathbf{J}_i |\Psi_{inv}\rangle = 0 \quad (\text{E.2})$$

where $\mathbf{J}_i$ is the total angular momentum operator acting on the i^{th} spin. This state has a physical interpretation in terms of the geometry of a single node of a spin-network state which has a single vertex and n edges attached to that vertex.

Dimension of Invariant Space

The dimension of the subspace consisting of invariant tensors, *i.e.* elements $|\Psi\rangle_{inv} \in \mathcal{H}_{inv}^n = \text{Inv} \left[\bigotimes_{i=1..n} \mathcal{H}^i \right]$ is given by the Verlinde formula [50]:

$$\dim(\mathcal{H}_{inv}^n) = \frac{2}{\pi} \int_0^\pi d\theta \sin^2(\theta/2) \prod_{i=1}^n \frac{\sin(j_i + 1/2)\theta}{\sin(\theta/2)} \quad (\text{E.3})$$